

# PAPER\_1505 — T\_SCm Activation Threshold = A\_5 = 60 K EXACT

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket L (LENR / SCm activation) **Date:** June 18, 2026 **Status:** CLOSED — SCm Heaviside activation temperature = icosahedral group order in kelvins

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## Observation

PAPER\_1072 (SCm Activation Function) introduces the Heaviside-smooth activation  $H_{SCm}(T) = 1/(1 + \exp(-(T - T_{SCm})/\Delta T))$  governing the SCm-channel switch-on. The activation threshold is specified as  $T_{SCm} \approx 60$  K, with reactor temperatures yielding  $H_{SCm} \approx 0.99$  (fully activated).

## UQFF Closed Identity

$T_{SCm} = A_5 \text{ K} = 60 \text{ K}$  EXACT

## Physical Interpretation

This is the **first identity equating an integer primitive directly to a physical temperature in SI units**. The icosahedral-group order  $A_5 = 60$  sets:

1. The SCm phonon condensation point at 60 K
2. The lower bound for LENR Heaviside activation
3. The same number that appears in Hayflick cellular limit (PAPER\_1373) and Pop III IMF (PAPER\_1373) — but here in kelvins

Below 60 K: SCm cluster formation suppressed, no LENR. Above 60 K:  $H_{SCm}$  rises sigmoidally toward 1.

$\omega_{SCm} = 1.25 \text{ THz}$  corresponds to  $\hbar\omega/k_B \approx 60 \text{ K}$ , providing thermodynamic consistency between phonon energy and activation temperature.

## NOT REPLACEMENT

SM treats LENR activation thresholds as empirical fit parameters per material system. UQFF supplies a single universal value rooted in the icosahedral integer primitive  $A_5$ .

## Reference

- Source: PAPER\_1072 SCm Activation Function
- Related: PAPER\_1373 ( $A_5$  in Hayflick/IMF), PAPER\_1141 (Rossi LENR variants),  $\omega_{SCm} = 1.25 \text{ THz}$
- Calculator dispatch: `calculate_paradox({"paradox": "t_scm_activation_threshold"})`

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